Dataset Report — RPG Wheat Parcels

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

FIELD LOCATIONS

Dataset role: Geographic anchors (where to extract features) · **Provider:** IGN / ASP (Agence de Services et de Paiement)
Download: region-segmented mirror, data.cquest.org · **Official portal:** geoservices.ign.fr/rpg
License: Licence Ouverte / Open Licence v2.0 (Etalab)

1. Source and provenance

1.1 What is the RPG?

The Registre Parcellaire Graphique (RPG) is France's official Geographic Information System for identifying agricultural parcels. It is the reference tool for processing and controlling Common Agricultural Policy (CAP) subsidies. Farmers declare the contours and the main crop of each parcel every year; an anonymised version is published openly. From the 2015 edition onward, crop information is provided at the individual parcel level (not just at the îlot level).

1.2 Where the data was retrieved

Item	Detail
Download source	data.cquest.org — RPG re-segmented by region (lighter than the national IGN files)
Region downloaded	Centre-Val de Loire (region code R24)
Editions	2020, 2021, 2022, 2023, 2024 (5 harvest seasons)
Projection	Lambert-93 (EPSG:2154)
Archive format	.7z (~150 MB each)

1.3 Format change in 2024

An important provenance detail: the RPG format changed in 2024.

Editions	Format	Layer used
2020–2023	ESRI Shapefile (RPG v2.0 / v2.2)	`PARCELLES_GRAPHIQUES.shp`
2024	GeoPackage (RPG v3.0, new 8-database offer)	`RPG_Parcelles.gpkg`

Despite the format change, all editions expose the same attributes, so they were processed uniformly. Both formats are read natively by GeoPandas.

2. Raw data characteristics

Attribute	Meaning
`ID_PARCEL`	Parcel identifier
`SURF_PARC`	Parcel area (hectares)
`CODE_CULTU`	Main crop code (soft winter wheat = "BTH")
`CODE_GROUP`	Crop group (1 = soft wheat)
`CULTURE_D1/D2`	Catch/secondary crops (mostly empty)
`geometry`	Parcel polygon

2.1 Wheat parcel counts (per year)

Year	Total parcels	Wheat parcels (BTH)	% wheat	Wheat area (ha)
2020	558,874	75,903	13.6%	552,484
2021	570,700	89,171	15.6%	639,667
2022	579,086	83,206	14.4%	605,195
2023	567,669	80,346	14.2%	604,269
2024	570,576	69,785	12.2%	546,311

Soft wheat consistently represents 12–16% of agricultural parcels — realistic for this leading cereal region, and stable across years, indicating clean data.

3. Transformations applied

Step 1 — Path resolution. Located the PARCELLES layer for each of the five editions across the nested folder structure (4 shapefiles + 1 GeoPackage).

Step 2 — Wheat filtering. For each year, kept only parcels where CODE_CULTU = "BTH" (soft winter wheat).

Step 3 — Size filtering. Removed parcels ≤ 1 ha to ensure a reliable satellite signal (a 10 m Sentinel-2 pixel needs several "pure" pixels per field).

Step 4 — Random sampling. Drew 300 parcels per year with a fixed random seed (42) for full reproducibility — 1,500 parcels total.

Step 5 — Centroid computation. Computed each parcel's centroid in Lambert-93 (metric, accurate), then converted to latitude/longitude (WGS84) for downstream APIs.

Step 6 — Unique identifier. Created `parcelle_uid` = year_IDPARCEL so each observation is uniquely traceable.

Step 7 — Export. Saved as CSV (for SoilGrids) and GeoJSON (for Google Earth Engine).

4. Key decisions and rationale

Decision: Sample wheat parcels independently for each year (not a fixed set across years).
Rationale: Crop rotation means a wheat field one year is usually a different crop the next. Per-year sampling guarantees every observed parcel was genuinely wheat that season. (Verified — see §5.)

Decision: Parcel-level granularity, 300 parcels/year (1,500 total).
Rationale: Parcel level produces enough rows for robust ML (vs ~130 at regional level) and captures spatial variability. 300/year balances dataset size against downstream extraction time (SoilGrids and GEE scale with the number of parcels).

Decision: Keep only parcels > 1 ha.
Rationale: Larger fields yield reliable satellite signals and avoid edge contamination, while a 1-ha (not 5-ha) threshold keeps the sample representative rather than biased toward only very large farms.

Decision: Fixed random seed (42).
Rationale: Makes the sample fully reproducible — supervisors can regenerate the exact same 1,500 parcels.

5. Validation — crop rotation verified

To confirm that per-year sampling is the correct design, the actual crop transitions between 2020 and 2021 were checked by matching parcels on their (stable) ID_PARCEL.

Result: Of 47,495 parcels in soft wheat in 2020, only **17.1% remained soft wheat in 2021**. The rest rotated to winter barley (29.3%), rapeseed (12.4%), spring barley (7.0%), sunflower (6.8%), maize (6.3%) and other crops. This is a textbook crop rotation and provides direct, data-driven justification for sampling wheat parcels separately each year.

Had a fixed set of parcels been reused across years, ~83% would not have been wheat in the following year, corrupting the dataset.

6. Resulting deliverable

Item	Value
Sample size	1,500 parcels (300 × 5 years)
Mean parcel size	7.9 ha (min 1.0, max 55.9)
Geographic control	Latitude 46.4–48.9, Longitude 0.1–3.1 — all within Centre-Val de Loire
Output files	`parcelles_ble_cvl_sample.csv` · `parcelles_ble_cvl_sample.geojson`
Key field	`parcelle_uid` (year_IDPARCEL)

7. Relevant observations

- **Data structure:** because parcels differ each year, the 1,500 rows are independent observations (a repeated cross-section), not a panel of 300 parcels tracked over time. This suits the prediction objective and adds spatial diversity.
- **Dual purpose:** this single parcel sample is the shared input for both the soil (SoilGrids) and satellite (Sentinel-2) feature extraction — the sampling was done once and reused.
- **Reusability:** the exported files are intended to be saved as a standalone Kaggle dataset so any downstream notebook can load them without re-sampling.

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